

Public Companies: Robotics & Humanoid Industry Landscape

Architecture, components, AI/technology, humanoid platforms, and manufacturing

1. Component Design (Actuators, Motors, Reducers, Sensors)

Company	Ticker	Role
Nidec	6594.T	Global incumbent in BLDC and coreless motors used across humanoid actuator stacks
Nabtesco	6268.T	Japanese maker of RV-series precision reduction gears adapted for humanoid hip/knee joints
Harmonic Drive Systems	6324.T	Leading maker of strain-wave (harmonic) reducers, the core joint-transmission component in humanoid limbs
Moog	MOG.A	Aerospace-grade actuator maker crossing into humanoid joint actuation
ROBOTIS	108490.KQ	Korean maker of Dynamixel servo motors, used widely in robotics research and expanding into reducers
Inovance Technology	300124.SZ	China's #1 servo motor maker; upstream supplier to Unitree and UBTech
Tuopu Group	601689.SS	Chinese supplier of rotary actuators, dexterous hands, and electronic skin for humanoids
Zhejiang Sanhua Intelligent Controls	002050.SZ	Thermal management components (valves, heat exchangers) for actuator and compute cooling
Sensata Technologies	ST	Sensor maker (force, position, temperature) supplying robotics and industrial customers
TE Connectivity	TEL	Connectors, sensors, and interconnects used in robot wiring harnesses and joints

2. Technology (AI Compute, Foundation Models, Perception)

Company	Ticker	Role
Nvidia	NVDA	Dominant robotics compute/software stack: Jetson Thor edge modules, Isaac GR00T vision-language-action models, Cosmos world models, Omniverse simulation
Qualcomm	QCOM	Dragonwing edge AI chips positioned as a robotics compute alternative to Nvidia
Intel	INTC	OpenVINO robotics toolkit and edge-AI silicon for robot perception workloads
AMD	AMD	GPU/compute supplier increasingly positioned for robotics and physical-AI training workloads

Mobileye	MBLY	Autonomy, perception, and edge-AI stack originally for vehicles, explicitly eyeing humanoid robotics as an adjacent market
Alphabet	GOOGL	DeepMind robotics research and cloud AI infrastructure underpinning third-party robot development

3. True Humanoid Robots (Public Platforms/Owners)

Company	Ticker	Role
Tesla	TSLA	Developing Optimus humanoid; piloting units in its own factories, targeting eventual public sale
UBTECH Robotics	9880.HK	Most direct pure-play humanoid stock; makes the Walker S full-size humanoid plus consumer/education robots
Hyundai Motor	005380.KS	Owns Boston Dynamics (Atlas humanoid, Spot, Stretch); piloting Atlas in its own Georgia manufacturing plant
Xiaomi	1810.HK	CyberOne/Iron humanoid platforms, positioned more as brand/technology demonstrations than near-term commercial products
Faraday Future	FFAI	Early-stage entrant reporting actual humanoid robot sales revenue (22 units shipped as of March 2026)
Richtech Robotics	RR	Small-cap service/hospitality robot maker with an early humanoid robot program (ADAM)

4. Manufacturing-Focused (Industrial Automation & Integration)

Company	Ticker	Role
ABB	ABBN.SW	Major industrial robot arm manufacturer; pursuing a robotics division spinoff to unlock pure-play value
FANUC	6954.T	Leading Japanese industrial robot and factory automation manufacturer
Yaskawa Electric	6506.T	Industrial robot and motion-control manufacturer, part of the ~2M-unit global industrial robot install base
Teradyne	TER	Owns Universal Robots (collaborative robot arms) and MiR (autonomous mobile robots)
Rockwell Automation	ROK	Programmable controllers, drives, and motion-control hardware; integrates industrial robots into production lines
Symbotic	SYM	Warehouse/logistics robotic automation systems manufacturer and integrator

Categories Worth Adding to Fully Characterize the Industry

The four buckets above cover the core stack, but a few distinct categories carry real investment weight and aren't captured cleanly by "component," "technology," "humanoid," or "manufacturing":

- **Materials & Rare Earth Supply Chain** — magnet and rare-earth suppliers (e.g. MP Materials) are a named bottleneck for actuator motors; this is upstream of "component design" and often analyzed separately.
- **Power & Energy Storage** — battery and power-density suppliers specific to mobile/humanoid platforms, distinct from stationary data-center power.
- **Simulation, Training Data & Testing Infrastructure** — e.g. Nvidia Omniverse/Isaac Sim, Hugging Face's LeRobot; increasingly treated as its own layer since robot learning depends on synthetic data/sim environments as much as chips.
- **Medical/Surgical Robotics** — a large, separate, and more mature public vertical (Intuitive Surgical, Globus Medical) that shares component suppliers but has entirely different end markets and economics.
- **Defense & Unmanned Systems** — ground/aerial autonomous systems (relevant given overlapping actuator and AI suppliers, different customer base and margin profile).
- **System Integrators / Deployment Services** — companies that install, program, and service robots at customer sites rather than manufacture them; a distinct margin and revenue-recognition profile from OEMs.

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